Color-vision deficiency screening

Chromatic families under simulated dichromacy

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1 Purpose

quiet-core states that its chromatic families have been screened under simulated protanopia, deuteranopia, and tritanopia, and that semantic redundancy, not hue distance, is the conformance mechanism.

This report reproduces that screening from the canonical palettes, for the chromatic anchors and for the line washes.

2 Method

Simulation uses the full-severity matrices of Machado, Oliveira, and Fernandes (2009), applied in linear sRGB. Simulated colors are analysis values: they are neither clipped nor gamut-checked.

Distances are Euclidean in OKLab, written ΔE. Both colors of a pair are simulated, as a viewer with the deficiency sees both.

quiet-core assigns no threshold of distinguishability to ΔE, and this report does not introduce one. The tables are for comparison: between pairs, and between normal and simulated vision.

3 Chromatic anchors

Table 1: Distance between chromatic anchors, quiet-day.

Pair	Normal	Protanopia	Deuteranopia	Tritanopia
Mineral / Moss	0.049	0.048	0.049	0.033
Mineral / Ochre	0.123	0.108	0.113	0.138
Mineral / Rust	0.124	0.084	0.094	0.160
Moss / Ochre	0.083	0.062	0.064	0.105
Moss / Rust	0.097	0.044	0.046	0.127
Ochre / Rust	0.039	0.030	0.020	0.023

Table 2: Distance between chromatic anchors, quiet-night.

Pair	Normal	Protanopia	Deuteranopia	Tritanopia
Mineral / Moss	0.049	0.048	0.050	0.034
Mineral / Ochre	0.111	0.093	0.110	0.117
Mineral / Rust	0.133	0.084	0.086	0.176
Moss / Ochre	0.071	0.046	0.064	0.085
Moss / Rust	0.111	0.054	0.037	0.142
Ochre / Rust	0.081	0.070	0.040	0.075

As expected for red–green deficiencies, the Moss and Rust pair loses the most distance under protanopia and deuteranopia. Under tritanopia, Mineral and Moss, and Ochre and Rust, draw closer while the other pairs separate.

In quiet-day, Ochre and Rust are the closest pair even with normal vision. quiet-core already forbids distinguishing warning from error by Ochre versus Rust alone.

Table 3: Distance between line washes, quiet-day.

Pair	Normal	Protanopia	Deuteranopia	Tritanopia
Mineral / Moss	0.023	0.023	0.023	0.012
Mineral / Ochre	0.038	0.032	0.034	0.046
Mineral / Rust	0.040	0.027	0.030	0.052
Moss / Ochre	0.025	0.012	0.011	0.034
Moss / Rust	0.032	0.012	0.007	0.040
Ochre / Rust	0.010	0.007	0.004	0.007

Table 4: Distance between line washes, quiet-night.

Pair	Normal	Protanopia	Deuteranopia	Tritanopia
Mineral / Moss	0.024	0.024	0.025	0.017
Mineral / Ochre	0.037	0.031	0.033	0.043
Mineral / Rust	0.040	0.024	0.026	0.054
Moss / Ochre	0.019	0.009	0.009	0.026
Moss / Rust	0.032	0.013	0.001	0.037
Ochre / Rust	0.016	0.011	0.007	0.012

4 Line washes

Washes are faint by design, so their distances are small in every condition. The consequential case is a diff: additions use the Moss wash and deletions the Rust wash.

Under deuteranopia those two washes nearly coincide in both adaptations. A reader with deuteranopia cannot tell an added line from a deleted line by its background.

This confirms the existing quiet-core rule rather than adding one: Wash supports meaning but must not be its sole carrier. Ports must keep the host's + and − markers, gutter signs, or equivalent cues for diff and diagnostic state.

5 Foreground contrast under simulation

WCAG contrast is defined on the colors actually displayed, not on a simulation. The table is therefore informative: it shows how the luminance a viewer perceives shifts under each deficiency.

Table 5: Chromatic anchors on Field, quiet-day.

Anchor on Field	Normal	Protanopia	Deuteranopia	Tritanopia
Mineral	4.59	4.34	4.77	4.57
Moss	4.60	4.47	4.69	4.61
Ochre	4.71	5.07	4.51	4.74
Rust	4.71	5.20	4.44	4.73

Because quiet-day's chromatic foregrounds sit close to 4.5:1, small simulated luminance shifts take some of them below it: Mineral and Moss under protanopia, and Rust under deuteranopia. quiet-night keeps every anchor above 4.5:1 under all three simulations.

These are not conformance failures. They are evidence for any future change-control proposal to raise quiet-day's chromatic contrast.

6 Limitations

Simulation is a screening tool. It does not replace evaluation with people who have color-vision deficiencies.

Only full-severity dichromacy is simulated. Anomalous trichromacy, which is more common, produces smaller shifts than those shown here.

Table 6: Chromatic anchors on Field, quiet-night.

Anchor on Field	Normal	Protanopia	Deuteranopia	Tritanopia
Mineral	5.55	5.78	5.41	5.57
Moss	5.58	5.65	5.52	5.55
Ochre	6.20	5.89	6.39	6.15
Rust	5.19	4.60	5.57	5.16